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Problem 1

Interpolation formula -> $Y = Y_1 + \frac{(Y_2 - Y_1)(x - x_1)}{(x_2 - x_1)}$

P = 0.01 MPa

T (°C)	v (m³/kg)	u (kJ/kg)
400	31.063	2969.3
450	33.372	3051.1
500	35.680	3132.9

$$V = 31.063 + \frac{(35.680 - 31.063)(450 - 400)}{(500 - 400)} = 33.372$$

$$u = 2969.3 + \frac{(3132.9 - 2969.3)(450 - 400)}{(500 - 400)} = 3051.1$$

P = 0.05 MPa

T (°C)	v (m³/kg)	u (kJ/kg)
400	6.2094	2968.9
450	6.6716	3050.75
500	7.1338	3132.6

$$V = 6.2094 + \frac{(7.1338 - 6.2094)(450 - 400)}{(500 - 400)} = 6.6716$$

$$u = 2968.9 + \frac{(3132.6 - 2968.9)(450 - 400)}{(500 - 400)} = 3050.75$$

Then interp at 450°C P = 0.03 MPa

P (MPa)	V (m³/kg)	u (kJ/kg)
0.01	33.372	3051.1
0.03	20.022	3050.93
0.05	6.6716	3050.75

$$V = 33.372 + \frac{(6.6716 - 33.372)(.03 - .01)}{(.05 - .01)} = 20.022$$

$$u = 3051.1 + \frac{(3050.75 - 3051.1)(.03 - .01)}{(.05 - .01)} = 3050.93$$